Massylia Mahdi

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Education

Bachelor of Applied Science – Computer Engineering *Sep 2022 – Apr 2026*
University of Ottawa, Ottawa, ON

Projects

Smart Baby Crib Monitoring System–Real-Time IoT Healthcare Device *Winter 2026*

- Designed and developed a smart crib system integrating an **STM32** microcontroller, environmental sensors, and a wearable monitoring bracelet
- Developed a smart bracelet prototype to track baby movement and heart rate using embedded sensors
- Integrated temperature sensing modules into the crib to monitor environmental conditions and detect abnormal variations
- Developed a user interface to display sensor data and alert generation through a buzzer
- Designed and modelled the smart crib enclosure, wearable monitoring bracelet, and sensor compartments using Fusion 360 and Tinkercad, then produced and validated functional prototypes through **3D printing**.

MediTrack Dispenser *Fall 2025 – Winter 2026*

- Developed features for a React Native mobile application (TypeScript) enabling doctors to manage prescriptions and patients to track medication adherence
- Implemented Firebase cloud services and resolved integration issues between mobile and embedded systems using **I2C, UART** connexion
- Designed system architecture integrating mobile app, cloud backend, and embedded controllers (ESP32, Raspberry Pi)
- Designed and modelled **3D CAD** enclosures and medication compartments using **Fusion 360 and Tinkercad**; fabricated functional prototypes through 3D printing to support embedded hardware integration, validation, and testing.

Real-Time Digital Voltmeter (STM32, FreeRTOS, C) *Fall 2025*

- Developed a real-time digital voltmeter on STM32 using FreeRTOS and C.
- Configured ADC, I2C, UART, and GPIO peripherals using STM32CubeIDE.
- Implemented a producer-consumer architecture with binary semaphores for task synchronization.
- Acquired 12-bit ADC measurements, converted analog signals to voltage values, and displayed results on an LCD via I2C.
- Applied real-time scheduling concepts, inter-task communication, and embedded firmware debugging.

Deep Learning for Health Prediction — Python, TensorFlow, Keras *Fall 2025*

- Developed and evaluated neural network models for binary classification using TensorFlow/Keras.
- Experimented with multiple architectures, activation functions, and hyperparameter configurations.
- Applied regularization techniques including Dropout, L2 regularization, and Early Stopping to improve model generalization.
- Evaluated model performance using Precision, Recall, F1-score, and ROC-AUC metrics.

Pipelined RISC Processor – VHDL *Winter 2026*

- Implemented a 5-stage pipelined RISC processor with hazard detection and forwarding units
- Verified functionality and timing using Quartus II and demonstrated execution on FPGA

Custom PC Service – Android Application *Fall 2024*

- Developed a Java mobile application with role-based authentication and SQLite database
- Designed UML diagrams (class, activity, sequence) to model system architecture

- Integrated UI animations (splash screen, transitions) to enhance user experience

Experience

Junior AI Analyst *July 2026 – Present*
Immigration and Refugee Board of Canada

- Designed and tested a bilingual (EN/FR) retrieval-augmented generation (RAG) chatbot in Microsoft Copilot Studio, including prompt engineering, agent instructions, and knowledge-grounding logic sourced from SharePoint and Dataverse.
- Designed and configured conversational topics, triggering phrases, and multi-turn dialogue flows in Microsoft Copilot Studio, including conditional branching logic and variable management for agent orchestration.
- Built two-pass retrieval orchestration logic (controlling topic + conditional fallback) to enforce knowledge-source precedence across multiple grounding sources.
- Experimented with multi-agent architectures (main agent + sub-agents) in Copilot Studio, evaluating trade-offs in model customization, variable passing, and orchestration.
- Conducted testing and quality assurance to evaluate AI Agents performance, identify issues, and improve solution reliability
- Collaborated with stakeholders to gather requirements, support ongoing projects, and contribute to process improvements

Customer Service Representative *Sep 2023 – Present*
City of Ottawa

- Analyzed client needs and recommended appropriate programs and services
- Managed participant records and program data using Excel spreadsheets to support camp registration, attendance tracking, and administrative operations
- Assisted with registrations and resolved technical and account-related issues

Teaching Assistant for Advanced Computer Systems Design course *Jan 2026 – April 2026*
University of Ottawa

- Assisted with labs and FPGA-based demonstrations (CPU, pipelining, memory hierarchy)
- Evaluated quizzes and midterms via Brightspace following department standards
- Provided technical support, debugging assistance, and guidance on software and hardware development concepts.

Tour Guide & Boutique *Summer 2023, 2025*
The Royal Canadian Mint

- Delivered bilingual tours tailored to diverse audiences
- Simplified complex production processes into clear and engaging explanations

Junior IT Analyst *Jan 2024 – Aug 2024*
Shared Services Canada

- Developed and maintained a SharePoint site with optimized user experience
- Leveraged Microsoft 365 tools (SharePoint, Teams, Outlook) to improve collaboration and workflow efficiency.
- Resolved technical issues and tracked tasks using Jira

Technical Skills & Certifications

- **Programming & tools:** Python(Pandas, NumPy, Scikit-learn), SQL, Java, C, HTML/CSS, JavaScript/TypeScript, Git, Linux, Android Studio, Jira, Microsoft Office(Excel, Power BI..), Cisco Packet Tracer, Wireshark, Copilot Studio
- **Technologies:** ReactJS, Node.js, Firebase, SQL, PostgreSQL, VHDL, Verilog (basic), Matlab, FreeRTOS
- **Security Clearance:** Government of Canada Reliability Status (Active)